

Zurich ServiceNow AI Platform Capabilities

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Company Headquarters

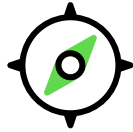
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CMDB Health

A healthy CMDB, based on health benchmarks in an organization, is essential to an effective and continuous use of the product. Health indicators such as duplicate CIs, required CI fields, and audits are evaluated and aggregated into health scores at the class, health group, and service levels. CMDB Health also provides a framework for applying standardized CI remediation.

Get started

Explore



Explore the KPIs and metrics that are evaluated for health, and the health results dashboards.

Configure



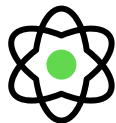
Select the health metric to monitor and configure any associated rules.

Use CMDB Health in CMDB Workspace



Access CMDB Health dashboards and related tasks in CMDB Workspace views.

Remediate



Create and apply remediation rules to CIs that failed health tests.

Track processes



Use logging and other troubleshooting methods to resolve issues with CMDB Health processes.

Reference



Details about properties, scheduled jobs, and tables, used by CMDB Health, and domain separation.

Troubleshoot and get help

- [Ask questions and explore other resources for in the ServiceNow Community](#)
- [Search the Known Error Portal for known error articles](#)
- [Contact Customer Service and Support](#)
- [Exploring CMDB Health](#)

Learn more about CMDB Health and review the benefits it can provide for users in your organization.

- [CMDB Health experience in CMDB Workspace](#)

View CMDB Health dashboards and use other CMDB Health-related features in CMDB Workspace.

- [View CMDB Health Dashboard](#)

The CMDB Health Dashboard serves as a central location to view detailed health reports for your CMDB at a glance. Also, it provides functions to remediate issues that CMDB Health detected, to improve CMDB health.

- [View relationships health](#)

View health scores for relationship health such as orphan, stale, and duplicate relationships, in the Relationship Health Dashboard.

- [View CI health](#)

View CMDB Health report for an individual CI, on the CIs form in CMDB Workspace.

- [Create a CMDB remediation rule](#)

A CMDB remediation rule is associated with a task that was created for a failed CMDB health test. A CMDB remediation rule is applied automatically or manually to execute a remediation workflow that can, for example, delete stale CIs.

- [CMDB Health process tracking and troubleshooting](#)

Use the following information to track and resolve issues with the CMDB Health processes.

- [CMDB Health reference](#)

Reference topics provide additional information about the KPIs and metrics that CMDB Health uses and about tables, scheduled jobs, and properties that are used by CMDB Health.

Exploring CMDB Health

Learn more about CMDB Health and review the benefits it can provide for users in your organization.

CMDB Health overview

CMDB Health Monitors and helps maintain the integrity and effectiveness of the CMDB, using health benchmarks specified for the organization. CMDB Health evaluates Health indicators such as duplicate CIs, required CI fields, and audits and aggregates them into health scores at the class, health group, and service levels. CMDB Health also provides a framework for applying standardized CI remediation for health issues that are detected.

CMDB Health benefits

CMDB Health is configurable so you can track specific areas of concern. It provides dashboards with aggregated numbers that you can easily glance and drill-down to locate the source of a problem.

What to explore next

To learn more about configuring and using CMDB Health, see:

- [Configuring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- View CMDB Health reports:
 - [View CMDB Health Dashboard](#)
 - [View relationships health](#)

- [View CI health](#)
- [Create a CMDB remediation rule](#)
- [Track CMDB Health processes](#)
- [CMDB Health reference](#)
- [Overview of CMDB Health](#)

Learn about the benefits, configuration, and usage of the CMDB Health feature.

- [CMDB Health KPIs and metrics](#)

CMDB Health calculates an aggregated score for three Key Performance Indicators (KPIs) which are correctness, compliance, and completeness, each further consisting of sub-metrics.

Related tasks

- [View relationships health](#)
- [View CI health](#)
- [Create a CMDB remediation rule](#)

Related concepts

- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

Overview of CMDB Health

Learn about the benefits, configuration, and usage of the CMDB Health feature.

The health of the CMDB data is monitored and reported for the following KPIs, each further consisting of sub-metrics:

Completeness

CI are tested for required and recommended fields that are not populated.

Correctness

CI are tested against predefined data integrity rules such as identification rules (to detect duplicate CIs), orphan CI rules, and stale CI rules.

Compliance

The CMDB data is audited for adherence to predefined certificates.

Relationships

The health of CI relationships is tested for indicators such as orphan and duplicate relationships, and for compliance with suggested relationships, hosting and containment rules.

After CIs are tested for various health indicators (metrics), the results are aggregated at the class, health group, service, and relationship level, and appear on CMDB Health dashboards. For most metrics, you can configure the health tests themselves to determine how compliance and non-compliance of CIs are evaluated.

For more details about the KPIs and their associated metrics, see [CMDB Health KPIs and metrics](#).

Note: CMDB Health doesn't support non-CMDB tables.

CMDB Health dashboards and reports

You can view CMDB Health dashboards and reports in [CMDB Workspace](#). Those dashboards show CMDB Health aggregated results for compliant and non-compliant CIs for CMDB Health KPIs and their metrics.

Dashboard/Report	Use
CMDB Health Dashboard 1. Navigate to Workspaces > CMDB Workspace. 2. Select Home in the CMDB Workspace menu bar. 3. In the Quick links section select the CMDB Health Dashboard link. or: 1. Navigate to All > Configuration > CMDB Health Dashboard. 2. Select one of the views.	Main CMDB Health Dashboard with the following views: Class View • Shows CIs and classes health reports with aggregations for metrics, KPIs, and class branches. • Access to remediation tasks that were generated for CIs that failed a health test. Service View • Shows service health reports. • Detailed health for CIs per service with aggregations for metrics and services. • Access to remediation tasks that were generated for CIs in a service that failed a health test. Health Group View • Shows health reports for CMDB groups of type Health. • Detailed health for CIs in the group, with aggregations for KPIs, metrics, and health groups. In either dashboard view you can select CMDB Health settings to manage the CMDB Health Dashboard jobs and configure other CMDB Health settings. For more information, see Configuring CMDB Health.

Dashboard/Report	Use
Relationship Health Dashboard 1. Navigate to Workspaces > CMDB Workspace. 2. Select Home in the CMDB Workspace menu bar. 3. In the Quick links section select the Relationship Health Dashboard link.	Shows various charts for relationships health, such as duplicate, orphan, and stale relationships. For more information, see View relationships health.
CI health report • On a CI form that isn't in CMDB Workspace, select Open in CMDB Workspace. • In CMDB Workspace, select a CI in a list view to open the CI's form. For example, on the CMDB Health Dashboard, drill down a KPI tile and then select a CI in the tile's list view.	On a CI form, the CI Health tile contains widgets that show CIs health reports. Select the right (>) or left (<) handles in the CI Health tile to scroll through the widgets.

Setup and configuration

To start gathering and aggregating health data, you must enable the CMDB Health-related jobs (CMDB Health Dashboard jobs) which are initially disabled. You also need to configure CMDB Health related system properties and health KPIs and metric test rules, to reflect health standards in your organization.

For details about setting up and configuring CMDB Health, see [Configuring CMDB Health](#).

Location	Configuration
CI Class Manager Configuration > CI Class Manager	Central location to manage CI classes and to configure CMDB health settings. Use the CI Class Manager to Manage rules and definitions that are used for health tests, such as orphan rules, audit certificates, and recommended fields rule. For information about configuring KPI and metric tests, see CMDB Health KPIs and metrics.
CMDB Health Preferences Navigate to Configuration > Health Preferences, or perform the following steps: 1. Navigate to All > Configuration > CMDB Health Dashboard. 2. Select a view. 3. On the CMDB Health Dashboard, select CMDB Health settings.	Central location for configuring CMDB Health settings: • Configure CMDB Health system properties. • Enable and manage the CMDB Health Dashboard jobs. • Activate or deactivate KPIs and metrics. • Set the maximum failure threshold for the KPIs and metrics. • Configure creation of tasks for failed CIs.

CI remediation

CMDB Health provides a framework for configuring [CI remediation](#). Remediation lets you proactively apply corrective actions to unhealthy CIs in a managed and standardized fashion.

Domain separation

CMDB Health is domain aware. If the domain separation plugin has been activated, then the CMDB Health Dashboard shows health based on data, rules, and settings from the logged-on user domain. If rules and settings aren't defined for a child domain, then the parent's settings are applied, recursively.

For more information, see [Domain separation in CMDB Health](#).

CMDB Health KPIs and metrics

CMDB Health calculates an aggregated score for three Key Performance Indicators (KPIs) which are correctness, compliance, and completeness, each further consisting of sub-metrics.

You can configure which KPIs and metrics are included in CMDB Health evaluations and dashboards. In the base system, all KPIs and all their metrics are included.

To access the CMDB Health Dashboard and view CMDB health, see [View CMDB Health Dashboard](#).

Correctness

A KPI which is an aggregation of the following metrics:

Orphan

Measures the percentage of orphan CIs in the CMDB. A CI can become orphan if it was unintentionally left in the CMDB when it is no longer needed. A CI is determined to be orphan if:

- The CI satisfies the criteria in an [orphan rule](#). This criteria checks for specific attributes that a CI must have, and for CIs that have no relationships or that don't have specific relationships.
- Data is missing for the CI in its respective table, or in one of its parents' table.

Staleness

Measures the percentage of stale CIs in the CMDB. A CI is stale if it was not updated within the **Effective Duration** time period that is specified in the [staleness rule](#) that applies to the class.

The base system includes a default staleness rule for the Configuration Item [cmdb_ci] class, which sets the **Effective Duration** time to 60 days. This rule applies to all extended CMDB classes, and can be overridden by class specific staleness rules defined by the user. To determine CI staleness, a staleness rule for the CI's class is used if it exists, otherwise, the default staleness rule is used.

In addition, a relationship in which a stale CI is a parent or a child, is determined to be a stale relationship.

Note:

Discovery marks VMware vCenter CIs that no longer physically exist, as stale. By default, this setting takes precedence over a CMDB Health staleness rule defined for the CI. When drilling-down in the CMDB Health Dashboard to Health Results, the **Source** for CIs determined to be stale by Discovery, is Cloud Discovery.

For the aggregated score of the correctness KPI to include only health results generated by CMDB Health, set the `glide.cmdb.health.src.cmdb_health_audit_only` system property to **true**. In which case, drilling-down to the details of stale CIs still shows those CIs in which source is Cloud Discovery. For more information, see [CMDB Health system properties](#) and [Discovery for VMware vCenter](#).

Duplicate

Measures the percentage of duplicate CIs in the CMDB using [identification rules](#). Only independent CIs are evaluated for duplication. In a set of duplicate CIs, the count of duplicate CIs is the total number of CIs in the set, minus one. The detailed graphs for a duplicate set of CIs display all the CIs in the set.

For more details, examples, and troubleshooting information about duplicate metric, see the [CMDB Health - Duplicate Metric - algorithm \[KB0726425\]](#) knowledge base article.

Completeness

A KPI which is an aggregation of the following metrics:

Required

Measures the percentage of CIs in which fields that are defined as mandatory, are not populated. Missing fields are tagged as incomplete noting that for this CI some information is missing. Required fields are equivalent to the fields that are [specified as mandatory](#) in the system dictionary.

Recommended

Measures the percentage of CIs in which fields that are [set as recommended](#), are not populated. Out-of-box, no recommended fields are specified.

You can use the [Add Identifier Fields In Recommended Rules](#) scheduled job to set criterion attributes from active identification rules, as recommended fields. You can use the [Remove Identifier Fields In Recommended Rules](#) scheduled job to unset criterion attributes from active identification rules, as recommended fields.

Compliance

Based on the results of actual CMDB audit runs.

Note: Audit must be activated for the compliance KPI to show results. For more information, see [Compliance Activation](#).

Audit

Audit compares actual values of specified fields, against expected values defined in template and scripted audits. Based on the Last run date of audits, CMDB Health identifies the set of the most recent complete audit run, and uses those audit results. To pass the CMDB Health audit test, a CI must be in compliance with all audits for that CI.

Create a compliance-type audit, for which the results are calculated into the CMDB Health compliance KPI.

When running [scripted audits](#), the Last run date is not populated. Therefore, for the compliance KPI to include the results of a scripted audit, update the script in the audit to record the audit run time.

Relationships

Measures the health of CI relationships, consisting of the following metrics which are not-configurable:

Duplicate relationships

Relationships that have identical parent and child CIs, identical relationship type, and an identical port. Duplicate relationships are displayed per relationship type. In a set of duplicate relationships, the duplicate relationship count is the total number of duplicate relationships in the set, minus one. The detailed graphs for a duplicate set of relationships display all the relationships in the set.

Orphan relationships

A relationship that is missing either a parent CI, a child CI, or both.

Stale relationships

A relationship in which the parent CI or the child CI is a stale CI.

A single relationship can fail more than one health test. For example, a duplicate relationship can also be stale.

Also reports the following relationship-related summaries:

- Relations not compliant with suggested relations
- Relations not compliant with containment rules
- Relations not compliant with hosting rules

To access the Relationship Health Dashboard and view relationships health, see [View relationships health](#).

Related tasks

- [Configure KPI and metrics preferences](#)

Related reference

- [CMDB Health system properties](#)

Configuring CMDB Health

Data collection for CMDB Health is highly configurable to reflect data health requirements and standards in your organization. Most importantly, the CMDB Health Dashboard jobs are disabled by default and health data isn't collected until you enable those jobs.

To report valuable and meaningful health data, review and adjust CMDB Health-related settings.

1. Review [CMDB Health KPIs and metrics](#) to learn what CMDB Health can monitor, and what to configure to enable and support each metric. Deactivate any KPIs and metrics that you aren't interested in reporting, and set failure thresholds.
2. For each KPI and associated metric that you want monitored, define the necessary rules and fulfill other needed requirements. For example, you might need to:
 - [Set a CI attribute to be mandatory](#)
 - [Set a CI field to be recommended](#)
 - [Create a CMDB Health orphan rule](#)
 - [Create a CMDB Health staleness rule](#)
3. Narrow the scope of CIs that are included in health calculations - see [Create health inclusion rule](#).
4. Examine and adjust as needed the [CMDB Health system properties](#).
5. Enable the Health Dashboard jobs for the KPIs that you want reported - for more information, see [Enable and configure a CMDB Health Dashboard job](#).

Configure KPI and metrics preferences

Include or exclude KPIs and metrics as needed in your organization, and configure system behaviour when CIs fail metric health tests.

Before you begin

To start collecting and reporting CMDB health KPIs and metrics, you must first [enable and configure the CMDB health dashboard jobs](#).

Role required: `itil_admin` (on top of `itil`)

About this task

If Domain Support - Domain Extensions is activated, then you can configure aggregation preferences per domain.

Procedure

1. Navigate to **All > Configuration > Health Preference**.
2. Select **Health Metrics** on the right-hand side bar.
 - a. In the **Select Metric** drop-down list select the **Completeness**, **Compliance**, or the **Correctness** KPI and toggle **Active** to included that KPI and its associated metrics in CMDB Health analysis and reporting.
 - b. In the **Select Metric** drop-down list select a metric, such as **Required**, and configure its associated fields.
For a metric:

Field	Description
Active	Activate the metric to include it in the aggregated health report for the respective KPI.
Failure Threshold	Threshold number of CIs that can fail the health metric test. When reached, health

Field	Description
	processing stops for the metric for this cycle. Maximum number is 100,000, even if you specify a higher number.
Create Task	If a CI fails the metric test, create a task with details about the failure. You can then view any generated tasks in the CMDB Workspace store app , and configure remediation for the task.
Task Assignee Group	An assignment group for the task.

3. Select **Save**.

Set a CI attribute to be mandatory

Configure a CI attribute as mandatory so it is included in the CMDB Health tests for the required metric if enabled. Required is a metric of the CMDB Health completeness KPI.

Before you begin

Role required:

- Itil_admin and personalize_dictionary: For editing the dictionary table
- itil has read access

About this task

When a field is configured as mandatory, then if the required metric is enabled, the CMDB health tests check whether that field is populated or not. The CMDB Health Dashboard shows the aggregated report of the percentage of CIs for which one or more required fields is empty.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to display the CI Classes list. Then select the class with the field that needs to be set as mandatory.
3. In the class navigation bar, expand **Class Info** and then select **Attributes**. In the Attributes view, click **Added**.
4. Locate the attribute that you want to set as mandatory, and then double-click its **Mandatory** value and set it to true.
The next time the form is opened, a field status indicator appears next to the field label, indicating that a value is mandatory.

Note: Mandatory fields are global. The field is marked as mandatory everywhere it appears on a form. Also, mandatory fields do not appear correctly when using Service Mapping tag-based discovery. For more information, see [Tag-based discovery in Service Mapping](#).

Related topics

- [CMDB Health Dashboard for Helsinki | Overview](#)
- [Make a field mandatory](#)

Set a CI field to be recommended

Define a list of CI fields as recommended, noting that it is desirable that they are populated by a data source such as Discovery. You can then configure the CMDB completeness KPI to include recommended fields in its aggregated health reports.

Before you begin

Role required:

- Itil_admin and personalize_dictionary: For editing the dictionary table
- itil has read access

About this task

Use this procedure for fields which should not be mandatory, but that might have useful information that the CI should have. For example, a field with information that might at some point help with diagnosis. Initially, a derived class is set with the recommended fields that are defined at the parent level. You can add or remove recommended fields for a derived class, setting it with its own recommended fields, without affecting the recommended fields at the parent or sibling levels. If all recommended fields for a derived class are removed, then the derived class automatically derives the recommended fields from its parent class.

The CI Class Manager doesn't support dot-walking for selecting fields from referenced tables, as recommended. See the steps at the end of this procedure to directly update tables with recommended fields, using one level dot-walking.

For more information about child and parent classes, see [Table extension and classes](#).

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to show the CI Classes list, and then select the class that contains the fields that need to be set as recommended.
3. In the class navigation bar, expand **Health**, select **Completeness**, and then select the Recommended Fields tab.
4. In the Recommended Fields tab, use the list collector to move the fields that you want to designate as recommended, from the **Available** list to the **Selected** list.
5. Click **Save**.

What to do next

To directly update the CMDB Recommended Fields [cmdb_recommended_fields] table, (outside the CI Class Manager), using dot-walking to designate a field from a referenced table, as recommended:

1. Select **All**.
2. In the Filter navigator, enter `cmdb_recommended_fields.list` to access the CMDB Recommended Fields table.
3. On the CMDB Recommended Field form, set **Table** to the table that you want to set with recommended fields.
4. Select **Recommended fields** and use the list collector to select recommended fields for the table. You can expand a reference field up to one level, and use dot-walking to select a field from a referenced table.
5. Select **Submit**.

Create a CMDB Health orphan rule

Create an orphan rule to determine the percentage of orphan CIs in the CMDB. This sum is then aggregated into the correctness CMDB Health KPI. Orphan rules are defined per class, and only a single orphan rule can be defined per class.

Before you begin

Role required: `itil` has read access, `itil_admin` (on top of `itil`) has full access.

About this task

Specify the conditions that CIs must meet to be considered an orphan CI. Specify attributes that a CI must have, relationships that a CI should not have, or both. In the relationship conditions, you can either specify that the CI has no relationships, or a set of specific relationships that the CI doesn't have.

Note: If there is a [health inclusion rule](#) for the orphan metric, then the conditions in the health inclusion rule and the conditions in the health orphan rule, shouldn't be identical.

A health orphan rule can for example, identify a CI of the `cmdb_ci_computer` class as an orphan CI if the CI is not set with an owner or an asset. Or, for example, you can create an orphan rule for a CI in the Disk [`cmdb_ci_disk`] class. If that CI doesn't have a

Contains::Contained by or Contained by::Contains relationship with a Computer CI, then it is an orphan CI.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to show the CI Classes list and then select the class for which to create an orphan rule.
3. In the class navigation bar, expand **Health**, select **Correctness**, and then select the Orphan Rule tab.
4. Select a rule to edit, if one exists, or select **New**, and then fill out the form.

Field	Description
Class	The class for which the orphan rule applies.
Attributes	Attribute conditions that a CI must satisfy to be considered an orphan CI. For example, the filter conditions in which both the Assigned to and the Owned by fields are empty, will identify the matching CIs as orphans.
Condition	And/Or operation between the Attributes conditions and the Relationship conditions.
Relationship	The relationship conditions that a CI must fail, based on records in the CI Relationship [cmdb_rel_ci] table, in order to be considered an orphan CI. To specify that a CI must have no relationships, choose

Field	Description
	Any Relation and Any Class respectively.

5. Select **Submit** or **Update** to save the rule.

Related topics

- [CMDB Health Dashboard for Helsinki | Overview](#)

Create a CMDB Health staleness rule

If the staleness metric is in effect, then staleness rules are used to determine the percentage of stale CIs in the CMDB. This sum is then aggregated into the correctness KPI.

Before you begin

Role required: itil has read access, itil_admin (on top of itil) has full access.

About this task

The Discovery setting of certain types of CIs as stale takes precedence over a CMDB Health staleness rule defined for the CI. For more information about Discovery marking CIs as stale, see [Discovery for VMware vCenter](#).

Staleness rules are defined per class. If a rule isn't defined for a class, then the parent's rule is automatically derived by the child class and applies for the child class. If later a staleness rule is defined at the child class level, it overwrites the staleness rule that was derived from the parent class.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to show the CI Classes list and then select the class for which to create a staleness rule.

3. In the class navigation bar, expand **Health**, select **Correctness**, and then select the Staleness Rule tab.
4. Select a staleness rule to edit or select **New**, and then fill out the Staleness Rule form.

Field	Description
Applies to	The class for which the rule applies.
Effective Duration	The time period that is used for the staleness test. If the CI was not updated (based on Updated sys_updated_on) within the specified time period — the CI is determined to be stale. If you enter a value with a prefix that is valid and a suffix that is not, such as 15 x — the valid portion of the value is used ('15'). If the entire value is invalid — the value is ignored and the previous valid value is used.

5. Select **Submit** or **Update** to save the rule.

Related topics

- [CMDB Health Dashboard for Helsinki | Overview](#)

Create health inclusion rule

Filter the CIs that are included in health calculations and that appear in the CMDB Health Dashboard by defining health inclusion rules. Use health inclusion rules to temporarily filter out classes that generate a large number of failures, until the problems are fixed.

Before you begin

Role required: itil has read access, itil_admin (on top of itil) has full access.

About this task

Evaluation for the required, orphan, recommended, duplicate and staleness health metrics, will apply only to CIs that satisfy health inclusion rules. For example, health inclusion rules can limit CMDB Health scores of the duplicate metric only for server and network CIs.

Note:

- You can specify health inclusion rules per domain.
- Creating health inclusion rules at the level of the base cmdb_ci table can potentially filter out health results of all classes in CMDB Health dashboards.
- Applying a health inclusion rule to the duplicate metric, is supported only in the global domain.
- Due to performance issues, dot-walking in health inclusion rules for the duplicate metric is not supported.

In addition to any health inclusion rules, [identification inclusion rules](#) also indirectly impact what appears in the CMDB Health Dashboard for duplicate CIs. The dashboard itself uses the identification engine (IRE) to identify duplicate CIs and therefore identification inclusion rules are applied.

Inheritance of health inclusion rules:

- If there are no health inclusion rules specified for a child class, then rules specified on a parent class are applied to the child class.
- If health inclusion rules are specified for a child class, then those rules take precedence over rules specified on a parent class.

In the base system, there are no predefined health inclusion rules, in which case all CIs are included in the CMDB Health calculations.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**.
2. Select **Hierarchy** to show the CI Classes list and then select the class for which to create a health inclusion rule.
3. In the class navigation bar, expand **Health** and then select **Health Inclusion Rules**.
4. Select an existing rule to edit or select **New** and then fill out the Health Inclusion Rules form.

Field	Description
Applies to	Class this rules applies to.
Active record condition	Criteria that CIs must meet to be included in the evaluation for the specified health metrics.
Applies to metric	Metrics that the rule applies to.

5. Select **Save** or **Update**.

What to do next

You can delete a health inclusion rules that is no longer needed by selecting that rule, and then on the CMDB Health Configuration form, selecting **Delete**.

Configure aggregation weights for CMDB Health scores

Metrics health scores are aggregated into their respective KPI (key performance indicator) scores, which in return are aggregated into the overall CMDB Health score. Modify the default aggregation weights for metrics and KPIs to reflect on the importance of one metric over another is assessing the health of CMDB in the organization.

About this task

The completeness KPI, for example, consists of the metrics required fields and recommended fields. By default, those metrics are configured to contribute 60% and 40% respectively to the aggregated score of the completeness KPI. You can modify that proportional weight of required fields and recommended fields within the completeness KPI to be, for example, 25% and 75% respectively. You can also modify the proportional weights of the completeness, compliance, and correctness KPIs within the aggregated overall score.

To modify aggregation weights, you must directly access the CMDB Health Metric Preferences [cmdb_health_metric_pref] table, where those settings are stored. However, custom settings are used only if the system is set to use legacy calculation methods.

If Domain Support - Domain Extensions is activated, then you can configure aggregation preferences per domain. By default, the weights of KPIs have default settings, and metrics are globally set.

Before you begin

Toggle the **Use legacy calculation methods** switch on the CMDB Health Dashboard to use legacy calculation methods. Custom weight settings are used in aggregated score calculations only if using legacy calculation methods is in effect.

Role required: itil or itil_admin

Procedure

1. Select **All** and then, in the navigation filter, enter `cmdb_health_metric_pref.list` to open the CMDB Health Metric Preferences table.
2. Select a metric, KPI, or the Overall entry, and set its Weighted average contribution attribute to the percentage number that you want.
Ensure that the sum of the percentage weights adds up to 100% for the respective aggregation:
 - The sum of the percentage weights of all metrics for a KPI, must add up to 100.

- The sum of the percentage weights of all three KPIs (correctness, compliance, completeness) must add up to 100.

CMDB Health system properties

Configure the following system properties to customize how CMDB Health monitors and evaluates CIs health.

Role required: `itil_admin`

Note: To open the System Property [sys_properties] table, enter `sys_properties.list` in the navigation filter.

Property	Description
Max time in minutes for which individual metric processor will run in each scheduled cycle [glide.cmdb.health.metricProcessor.maxRunningTime]	If processing of a metric exceeds the specified time, CMDB Health processing halts until the next CMDB Health job is scheduled to run. • Type: integer • Default value: 120 • Location: Navigate to All > Configuration > Health Preference. In the right hand-side pane, select System Properties. For performance reasons, it is recommended not to set this property to a value greater than 120. Note: If you enter an invalid value, the default value is used.

Property	Description
Attribute in the [cmdb_ci] table that references the User or User Group table. CMDB Health Dashboard charts use this attribute to determine CIs' ownership. glide.cmdb.health.ci_ownership_field	Attribute in the [cmdb_ci] table that the CMDB Health Dashboard uses to determine CI ownership. - Type: string - Default value: managed_by_group - Location: Navigate to All > Configuration > Health Preference. In the right hand-side pane, select System Properties.
glide.cmdb.logger.use_syslog.CMDBHealth	A comma-separated list that controls the level of logging of CMDB Health jobs. Logging creates entries in the system logs to capture messages generated by the health auditing process each time they run. This logging helps debugging if there is a failure. For example, to log error and info messages, set the value to 'error,info'. - Type: String - Default value: error - Other possible values: Comma-separated list with any of the following values: - info - error - warn

Property	Description
	Or '*' which is equivalent to including all possible values. Location: System Property [sys_properties] table.
glide.cmdb.health.src.cmdb_health_audit_only	When set to true, disables health results from sources other than CMDB Health audit (such as Cloud Discovery). Only results generated by CMDB Health audit appear for KPIs in the CMDB Health Dashboard. For example, by default, if a CI is determined to be stale by Discovery, then that CI is counted as stale for the correctness KPI even though CMDB Health audit didn't determine that CI to be stale. However, in this case, drilling-down the stale CIs metric, still shows those CIs in which source is Cloud Discovery. - Type: true false - Default value: false - Location: System Property [sys_properties] table. - Learn more: CMDB Health KPIs and metrics Discovery for VMware vCenter

Property	Description
glide.cmdb.health.staleness_exclude_dependent_cis	Exclude dependent CIs for the staleness CMDB Health metric. When enabled, dependent CIs are not checked for staleness, regardless of any staleness or inclusion rules that are defined for the respective CI types. • Type: true false • Default value: false • Location: System Property [sys_properties] table.

Enable and configure a CMDB Health Dashboard job

Enable and configure the CMDB Health Dashboard jobs that process CMDB Health tests, to start calculating CMDB Health scores for the completeness, compliance, correctness KPIs, associated metrics, and relationships.

Before you begin

Role required: admin

About this task

In the base system, CMDB Health Dashboard jobs are disabled by default. Enable and configure the respective job for the CMDB health KPI that you want data collected and aggregated for. You can schedule a job to run on a recurring schedule, or run it once at any time.

For more information about how CMDB Health Dashboard jobs work, see the [Understanding the CMDB HealthDashboard Numbers](#) blog post in the ServiceNow Community.

Procedure

1. Navigate to **All > Configuration > Health Preference**.
2. Select **Scheduled Jobs** on the left-side bar and then select the job that you want to enable or configure.

CMDB Health Dashboard job	Description
CMDB Health Dashboard - Completeness Score Calculation	Script for calculating the completeness KPI of CMDB health.
CMDB Health Dashboard - Compliance Score Calculation	Script for calculating the compliance KPI of CMDB health.
CMDB Health Dashboard - Correctness Score Calculation	Script for calculating the correctness KPI of CMDB health.
CMDB Health Dashboard - Relationship Score Calculation	Script for calculating the CI relationships KPI of CMDB health.
CMDB Health Dashboard - Relationship Compliance Processor	Script for calculating compliance of relationships with suggested relationships, and with hosting and containment rules.

3. Review the default configuration, and update as necessary.

Field	Description
Name	Job name. Leave this field with the pre-populated name.
Active	Select to activate the job.
Conditional	If selected, the scripted condition must evaluate to true before the job can run.

Field	Description
Run	Configure the schedule for job execution, or select On Demand to run the job manually when needed. If Active is selected, then additional fields appear according to your choice. For example, Repeat Interval (when Run is set to Periodically), and Time for various settings of Run. Fill in the details to set precise running times.
Time zone	Local time zone.
Run this script	The job's script. Note: Changes to the script might result in unexpected behavior.

- (Optional) Select **Execute Now** to immediately run the job once.

Result

After you enable a CMDB Health Dashboard job, the results for the KPI are aggregated and appear in the CMDB Health dashboard and CI Health widget on CI forms, at the CMDB, class, and CI levels.

CMDB Health experience in CMDB Workspace

View CMDB Health dashboards and use other CMDB Health-related features in CMDB Workspace.

Starting with [CMDB Workspace](#) version 7.0, CMDB Health is fully integrated into CMDB Workspace. CMDB Workspace is required in order to access CMDB Health features, primarily to:

- [View CMDB Health Dashboard](#)
- [View relationships health](#)
- [View CI health](#)

In addition, you can use the CMDB Workspace landing page and its views to access CMDB Health files that are based on CMDB Health activities and aggregations. For example:

- Use the Important actions tile on the landing page to access cards with CMDB Health-related tasks such as de-duplication tasks.
- Use the CMDB Health tile to see overall health scores for the completeness, compliance, and correctness KPIs, and to drill down to the CMDB Health Dashboard within CMDB Workspace.
- Use tiles throughout the CMDB Workspace to drill down to health overviews of specific CIs.

Related tasks

- [View relationships health](#)
- [View CI health](#)
- [Create a CMDB remediation rule](#)

Related concepts

- [Exploring CMDB Health](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

Related reference

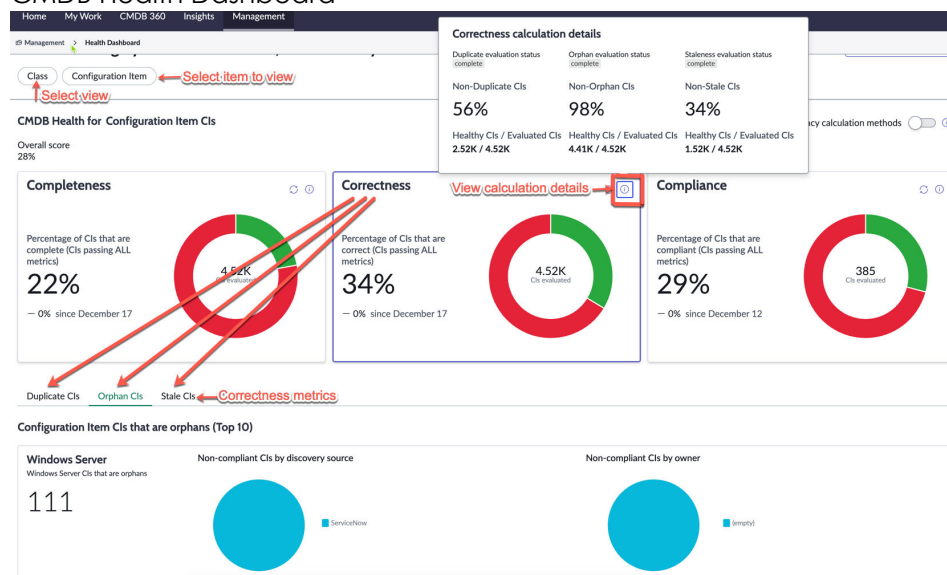
- [CMDB Health process tracking and troubleshooting](#)

View CMDB Health Dashboard

The CMDB Health Dashboard serves as a central location to view detailed health reports for your CMDB at a glance. Also, it provides functions to remediate issues that CMDB Health detected, to improve CMDB health.

Once you configure CMDB Health and the CMDB Health Dashboard Jobs are enabled, the CMDB Health Dashboard shows meaningful data that is automatically collected and calculated on a recurring schedule. For more information, see [Configuring CMDB Health](#).

CMDB Health Dashboard



Access

Use either of the following ways to access the CMDB Health Dashboard:

Within CMDB Workspace

1. Navigate to **Workspaces > CMDB Workspace**.
2. Select **Home** in the CMDB Workspace menu bar and then, in the Quick links section, select the **CMDB Health Dashboard** link.

Navigation filter

Navigate to **All > Configuration > CMDB Health Dashboard** and then select one of the following dashboard views:

- **Class View:** Default view for the CMDB Health Dashboard, showing health reports for CIs and classes in the CMDB hierarchy.
- **Service View:** Health reports for services with details for CIs per service.
- **Health Group View:** Health reports for CMDB groups of type Health with details for CIs per group.

Role requirement: Using the CMDB Health Dashboard requires the asset, itil, or cmdb_user role.

Legacy calculation methods

By default, all metrics and KPIs are set with a specific percentage weight in score calculations. However, you can use legacy calculation methods which were used up until the Washington DC release by toggling the **Use legacy calculation methods** switch (requires the sn_cmdb_admin role) on the CMDB Health Dashboard. In those methods, calculating aggregation scores is based on weights settings for metrics within their KPI, and KPIs within the overall score. Use the legacy calculation methods when it is important to reflect on specific weights of metrics and KPIs in the assessment of CMDB health in your organization.

For more information about customizing calculation weights for metrics, KPIs, and the overall score, see [Configure aggregation weights for CMDB Health scores](#).

KPI tiles

The Completeness, Correctness, and Compliance KPI tiles on the dashboard, show the aggregated health for CIs of the specified class (and its descendants), health group, or service. The KPI's aggregated percentage is represented on the donut as follows:

- Green: Represents the CIs that are compliant on all of the KPI's metrics
- Red: Represents the CIs that fail one or more metric

The distribution of metric compliance can vary greatly across CIs. For example, a single CI failing all metrics or 3 unique CIs each failing a different metric will result in a different KPI final score of aggregation. You can show additional details and further drill down the CMDB Health Dashboard:

- Point to the green and red portions in the donut reports, to show details.
- Select the View calculation details icon ("i") in a KPI tile to show the calculation details of failure and success for each of the KPI metrics. Calculation details also show the status of the calculation, such as complete.

•

Select a KPI tile and then select a tab for a metric that you want to see more details for. For each metric, the dashboard shows up to ten classes (each class in its own tile) with the highest number of non-compliant CIs for the metric. Each tile for a class shows the total number of non-compliant CIs and also pie reports of those non-compliant CIs by discovery source and by owner.

For example:

- 1: Select the Correctness tile.
- 2: Select the Orphan CIs tab.
- 3: Assuming that the largest number of CIs that aren't compliant with the orphan metric are Windows Servers, the first tile shows details for those CIs. The tile shows the total number of Windows Server CIs that are orphans. This tile also shows pie reports of those CIs, one by discovery source and another by owner.
- 4: Select a CI in the list view to open its CI form.
- 5: Scroll in the CI Health tile to the CMDB Health widgets with details about the CI's health.

The '<x>% since <date>' shows the increase or decrease (in %) in the KPI health since <date> (<date> being the previous time that change

was noted). A  icon indicates that the maximum failure threshold for the metric has been reached. The tests for the metric are halted for this cycle, and all associated aggregated summaries show 0%.

Overall score

The overall score appears on the dashboard above the KPI tiles and represents the aggregated health score for the CMDB. The overall score is calculated as an aggregation of the three KPIs (correctness, completeness and compliance). By default, the overall score is a simple average of the KPI scores. However, if using the legacy calculation methods is in effect, then the overall score is based on any weight settings of any of the KPIs.

Class view in the CMDB Health Dashboard

The class view shows health reports for CIs and classes in the CMDB hierarchy. Select any class in the CMDB hierarchy to filter all tiles on the dashboard to show health data only for CIs in the selected class and its child classes.

Select the **Class** drop-down to switch to the **Health Group** or the **Service** view.

Health Group view in the CMDB Health Dashboard

Select the **Health Group** view to show health reports for CIs of a CMDB health group. Select any health group to filter all tiles on the dashboard to show health data only for CIs in the selected health group.

Select the **Health Group** drop-down to switch to the **Class** or the **Service** view.

Service view in the CMDB Health Dashboard

Select the **Service** view to show health reports for CIs of a service class, such as Business Service, including the service CI itself. Select any service to filter all tiles on the dashboard to show health data only for CIs in the selected service.

Note: The Event Management and Service Mapping Core plugin must be activated to enable meaningful data in the service view.

Select the **Service** drop-down to switch to the **Health Group** or the **Class** view.

Settings affecting the service view:

- The service view uses the settings for the Business Service, Manual Service, and Technology Management Service classes.
- The service view doesn't include all services from the Service [cmdb_ci_service] table. Only services from the cmdb_ci_service_auto table and its descendants (cmdb_ci_service_discovered, cmdb_ci_service_manual, cmdb_ci_query_based_service), are included.
- The [glide.cmdb.services_hierarchy_limit](#) system property limits the number of service CIs that appear in the service view. This limit applies to any child class of the Service Instance [cmdb_ci_service_auto] class and is set to 10,000 by default.

Related tasks

- [View relationships health](#)
- [View CI health](#)
- [Create a CMDB remediation rule](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [CMDB Health reference](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

View relationships health

View health scores for relationship health such as orphan, stale, and duplicate relationships, in the Relationship Health Dashboard.

Before you begin

The CMDB Health Dashboard - Relationship Compliance Processor dashboard job must run to generate data for these reports.

Note: As a general guideline, activate this job only if you need the relationship reports as this job can affect performance.

Role required: itil or asset

About this task

The Relationship Health Dashboard shows various scorecards for health indicators of CI relationships in your CMDB. It contains donut reports for duplicate, orphan or stale relationships, broken down by relationship type. You can drill down these charts for further details.

The  icon indicates that the maximum failure threshold for the metric has been reached. The tests for the metric are halted for this cycle, and all associated aggregated summaries show 0%. Review the scorecard rules which might be ineffective, or the CMDB might be in an unstable state.

CMDB Health reports for the following metrics:

Duplicate relationship

Relationships that have identical parent, child, and relationship type.

Orphan relationship

A relationship that is missing parent, child, or relationship type.

Stale relationship

A relationship in which one of the CIs is stale. For a stale CI — its associated relationships are also stale.

In addition, the following relationship compliance reports with breakdown by relationship types, are available:

Relationships not compliant with all relationship rules

Relationships that do not comply with any [relationship governance](#) rule, including suggested relationships and dependent relationship rules.

Relationships not compliant with suggested relationships

[Suggested CI relationships](#) are used as rules to test if relationships comply with specified suggested relationships.

Relationships not compliant with containment rules

Containment rules are used to test if relationships comply with specified containment relationships.

Relationships not compliant with hosting rules

Hosting rules are used to test if relationships comply with specified hosting relationships.

For each of the compliance reports, testing a relationship requires a rule (suggested relationship, hosting rule, or containment rule) in which the parent and child CI classes match the parent and child CI classes in the tested relationship. If the relationship types in the rule and in the tested relationship don't match, then the relationship isn't in compliance. If an applicable rule isn't found, then the relationship is considered to be in compliance. Rules apply to the classes specified in the rule, and also to descendant classes. Therefore, when testing a relationship, rules that apply to ascendant parents of the CIs in the tested relationship are used. If there are multiple rules that match the parent and the child CI classes of the tested relationship, then the tested relationship needs to satisfy only one of these rules to be in compliance.

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. Select **Home** in the CMDB Workspace menu bar and then, in the Quick links section select the **Relationship Health Dashboard** link.
3. Examine the donut reports and the charts in the Relationship health overview and Relationship compliance overview sections on the dashboard.
Report results are grouped by relationship type, and you can drill down for further details:

- Point to a donut report to show more details about compliant and non-compliant relationships.
- Point to a bar in a bar chart to show more details for the bar.
- Select a bar in a bar chart to drill down to a list view of all the associated relationships that aren't compliant. The list view includes the **Failure Description** field which lists only a single rule that the relationship didn't comply with, even if there are additional rules that the relationship fails to comply with.

What to do next

For troubleshooting information, see the [How to identify and delete duplicate CMDB CI Relationship records, or ones that have orphan or missing parent/child relationships \[KB0780988\]](#) knowledge base article.

Related tasks

- [View CI health](#)
- [Create a CMDB remediation rule](#)
- [Enable and configure a CMDB Health Dashboard job](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

View CI health

View CMDB Health report for an individual CI, on the CIs form in CMDB Workspace.

Before you begin

Starting with [CMDB Workspace](#) version 7.0, CMDB Health is fully integrated into CMDB Workspace and is required in order to access CMDB Health features, including to view CI health.

Role required: sn_cmdb_user (CMDB user)

About this task

CI health scores are based on settings of CMDB Health KPIs and metrics. Aggregations are calculated in real-time from data stored in health-related tables which the CMDB Health Dashboard jobs update on a recurring schedule. The completeness and correctness KPIs are always up to date. However, the compliance KPI depends on audit cycles and on the CMDB Health Dashboard - Compliance Score Calculation dashboard job that might not have run yet.

To ensure that the latest updates to these KPIs are reflected, navigate to the respective CMDB Health dashboard job, and select **Execute Now**. For more information, see [Enable and configure a CMDB Health Dashboard job](#).

Procedure

1. Open the CI form for a CI, in CMDB Workspace.

-

On a CI form that isn't in CMDB Workspace, select **Open in CMDB Workspace**.

Role required: sn_cmdb_user (minimum)

- In CMDB Workspace, select a CI in a list view to open the CI's form. For example, on the CMDB Health Dashboard, drill down a KPI tile and then select a CI in the tile's list view.
2. Select the right (>) or left (<) handles in the CI Health tile to see all CI health-related widgets.

Result

The widgets in the CI Health tile on the CI form show the aggregated health for the CI using the following color codes:

- Green: The CI passed the health test (for example, it isn't a duplicate).
- Red: The CI failed the health test (for example, it is a duplicate)
- Grey: The CI wasn't tested for this metric, because the threshold wasn't set for the CI (class) in the CI module.

Note: Missing rules or other class definitions can prevent some health scores from being evaluated for a CI. The results in the CI Health widget in these situations, are described below:

Duplicate

- If no identification rules ([cmdb_identifier]) are defined for the CI's class or its ancestors: A notification to that effect appears.
- If only dependent identification rules are defined: Not applicable notification appears.

Orphan

- If the CI is excluded by health inclusion rules: Not applicable notification appears.
- If no orphan rules (in the [cmdb_health_orphan_rule] table) are defined for the CI's class or its ancestors: A notification about missing a rule appears.

Staleness

- If the CI is excluded by health inclusion rules: Not applicable notification appears.
- If no staleness rules (in the [cmdb_health_staleness_rule] table) are defined for the CI's class or its ancestors: A notification about missing a rule appears.

Audit

- If no audits ([cert_audit]) are defined for the CI (The CI Health widget checks only desired states and scripted audits): Not applicable notification appears.
- If there are audits defined for the CI but the audits did not run: Not applicable notification appears.

Related tasks

- [View relationships health](#)

- [Create a CMDB remediation rule](#)
- [Enable and configure a CMDB Health Dashboard job](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

Related topics

- [Timeline of CI changes](#)

Create a CMDB remediation rule

A CMDB remediation rule is associated with a task that was created for a failed CMDB health test. A CMDB remediation rule is applied automatically or manually to execute a remediation workflow that can, for example, delete stale CIs.

Before you begin

You must first create and publish a remediation workflow that addresses the CI issue, stored in the Workflow [wf_workflow] table. The workflow can be a regular workflow or an Orchestration workflow, and must adhere to the following requirements:

- The table in the workflow must match the task type in the remediation rule.
- The workflow isn't configured with any filter conditions by setting **If condition matches** to None, so that the filters of the CMDB remediation rule will apply.

For more information about using Classic Workflows, see [Classic Workflow](#).

Role required: itil_admin (on top of itil)

Procedure

1. Navigate to **All > Configuration > CMDB Remediations**.
2. Select **New** and then fill out the CMDB Remediation Rule form.

Field	Description
Name	Remediation name.
Task type	Type of CMDB health-related tasks to apply the remediation to.
Task filter	Filters tasks to apply remediation to. Also applies dot-walking on CI fields so that remediation is applied to tasks associated with matching CIs.
Execution	• Manual: Remediation is applied manually. • Automatic: The workflow is applied once, upon the creation of a task that matches the Task type and Task filter.
Active	Allowing the workflow to run.
Workflow	The CMDB remediation workflow (regular or Orchestration) that will execute automatically or manually, depending on the Execution setting. You can click the Lookup using list icon, and

Field	Description
	then click New to create a new workflow.

3. Select **Submit**.

Result

If **Execution** is set to Automatic, then the business rule Run remediations for CMDBHealth task applies the remediation workflow to CIs that match the Task filter. If **Execution** is set to Manual, then you can manually apply the remediation workflow defined in the rule.

- [Apply CMDB remediation](#)

Manually initiate a workflow to remediate a CI that failed a CMDB health test. For example, you can remediate CIs that are orphan or stale.

Related tasks

- [View relationships health](#)
- [View CI health](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

Related topics

- [Dot-Walking](#)

Apply CMDB remediation

Manually initiate a workflow to remediate a CI that failed a CMDB health test. For example, you can remediate CIs that are orphan or stale.

Before you begin

Role required: `itil_admin`

To manually apply a CMDB remediation, a CMDB remediation rule must exist, in which **Execution** is set to **Manual**.

About this task

Except for the duplicate and audit health metrics, you can choose to create tasks for health test failures for a metric.

To remediate failures of the duplicate metric, use [de-duplication tasks](#).

For all metrics except for audit, each CI that failed a metric test is associated with a single task. Because a CI can fail multiple audits, a single CI can be associated with multiple audit tasks. The first of those tasks is in the **Task** field, and any additional tasks are in the **Additional Tasks** field. To remediate failures of the audit metric, refer to the audit tasks for the audits that the CI failed.

CMDB remediation is applied in the [CMDB Workspace store app](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.
2. Select **Home** in the CMDB Workspace menu bar and then, in the Quick links section, select the **CMDB Health Dashboard** link. Ensure that the CMDB Health Dashboard is in **Class** view.
3. Select one of the KPI tiles (Completeness, Correctness, or Compliance).

4. Select the metric tab that is associated with the remediation that you want to apply (such as Stale CIs) and then select one of the class files.
5. In the list view of the non-compliant CIs for the metric, select the task link (in the Task column) for a CI that you want to remediate.
6. On the remediation form, select **Remediate**.
7. In the Remediate dialog box, select the remediation workflow that you want to apply.
The list of remediation workflows is based on the type of health metric (such as orphan, stale), and on the filter defined for the workflow in the associated remediation rule.
8. Select **Next**.

Related tasks

- [Create a CMDB remediation rule](#)

CMDB Health process tracking and troubleshooting

Use the following information to track and resolve issues with the CMDB Health processes.

Logging

By default, only error messages are logged to the syslog table, with the source name CmdbHealth. To enable logging of 'info' and 'warning' messages (which are typically logged at the start and end of each processing cycle), update the system property glide.cmdb.logger.use_syslog.CMDBHealth. For information about using this property, see [CMDB Health system properties](#).

Processing status

If scheduled jobs are enabled, but data is not displaying on the [var.config-mgmt-database-short](#) dashboard, you can check the processing status in the CMDB Health Metric Status [cmdb_health_metric_status] table. Depending on the status of the inactive metric, decide how to proceed.

Initially, the state of all metrics is 'In Progress'.

Possible final states of a metric:

Complete

All classes are processed and the number of failures is under the maximum failures threshold.

Max Failures

The number of failures for this metric reached the maximum failures threshold. Processing has been aborted and will start over in the next run.

Daily Time Out Pause

The processor reached the processing time limit. Processing is paused and will resume in the next run.

At the end of a processing cycle, the final state of a KPI depends on the final state of its associated metrics. Possible final state of a KPI:

Complete

All associated metrics are in Complete state and score calculation is complete.

Incomplete

Score is not calculated because one of the associated metrics reached its maximum failure thresholds.

Daily Time Out Pause

Timed out because one of the associated metrics has reached its processing time limit.

Processing time

If processing of a metric times out, you can find out which class takes too long to process. Use this information to find out if any validation rules are weak.

The progress of each metric is tracked in the CMDB Health Processor Status table [cmdb_health_processor_status]. Status for classes that have

been processed for a metric is Complete, and for classes that are yet to be processed is Draft. By looking at the update time for each class, you can calculate the length of processing time for each class.

Orphan records due to broken hierarchy

Orphan rules might detect an orphan CI, which you are not able to access and delete. Or, there might be a mismatch between the list view that displays the orphan records, and the total number of records. These findings are due to records being deleted in the database from only one table in the CMDB hierarchy.

These CI records are not accessible via GlideRecord and must be deleted directly from the database. Therefore, in this case, to delete an orphan CI from the database you must contact Support to get help.

Orphan test results provide the details of where exactly the hierarchy is broken. For example, the message "This cmdb_ci_linux_server CI [91054fc24f22520053d6e1d18110c713] is missing record in cmdb_ci_computer table" means that a record of that sys_id must be deleted from the CMDB, cmdb_ci, cmdb_ci_hardware, cmdb_ci_server, and the cmdb_ci_linux_server tables (the Computer class is between the Hardware and the Server classes in the hierarchy.)

Scripted audits Skipped

An error message is logged if the results from a scripted audit are not included in the compliance KPI. The reason can be that the script in the audit was not updated to populate its **Last ran date** field. Without a **Last ran date** value, CMDB Health is unable to identify these run results as part of a recent complete audit run, and skips those results.

- [CMDB Health process status: failure threshold reached](#)

The CMDB Health Dashboard shows the string 'failure threshold reached' when the number of CIs that are failing the metric tests, reaches the failure threshold set for the metric.

- [CMDB Health process status: incomplete score](#)

The CMDB Health Dashboard shows the string 'incomplete score' for a metric when it fails to calculate the score for the metric.

Related tasks

- [View relationships health](#)
- [View CI health](#)
- [Create a CMDB remediation rule](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)
- [CMDB Health reference](#)

CMDB Health process status: failure threshold reached

The CMDB Health Dashboard shows the string 'failure threshold reached' when the number of CIs that are failing the metric tests, reaches the failure threshold set for the metric.

CMDB Health stops processing for this metric in the current cycle, and therefore there is no aggregated health score for the metric. Processing will be attempted again in the next cycle. Also, status in the CMDB Health Metric Status [cmdb_health_metric_status] table is set to Max Failures for this metric.

When the health score of a metric cannot be evaluated, then the processing status of the respective KPI (for example, correctness) is set to Incomplete. The CMDB Health Dashboard shows the string Incomplete score for the respective KPI. Also, aggregated health scores for the metric are not available for any class in the CMDB hierarchy.

Review and refine the rules defined for the metric which has reached max failures. If a rule associated with the metric is too generic, resulting in large number of failures, attempt to refine it as follows:

-

Completeness:

Review the mandatory fields for the associated classes and remove those that aren't critical for the health score. Also, review the recommended fields that are causing failures and remove those that aren't critical for the health score.

For more information see [Set a CI attribute to be mandatory](#) and [Set a CI field to be recommended](#).

-

Compliance:

Review the audit failures from the audit jobs and adjust the audits to reduce the number of failures. For more information, see [CMDB Health KPIs and metrics](#).

-

Correctness:

- Orphan: Review the orphan rules and adjust them to remove excessive orphan failures. For more information, see [Create a CMDB Health orphan rule](#).
- Staleness: Review the effective duration for the classes that are causing failures and adjust the duration to reduce the number of failures. For more information, see [Create a CMDB Health staleness rule](#).
- Duplicate: Review the de-duplication tasks and remediate the tasks to remove duplicate CIs and to avoid creating failure records. For more information, see [Remediate a de-duplication task \(manual\)](#).

For more troubleshooting information about failure threshold related issues, see [How to Resolve Max Failure Threshold Errors for CMDB Health Dashboard \[KB2487349\]](#).

CMDB Health process status: incomplete score

The CMDB Health Dashboard shows the string 'incomplete score' for a metric when it fails to calculate the score for the metric.

'Incomplete score' is displayed when:

- The number of CIs that are failing the tests of one of its sub-metric, reaches the failure threshold set for the metric. In this situation, the processing status for the respective parent metric (for example, correctness) is set to 'incomplete' in the CMDB Health Metric Status [cmdb_health_metric_status] table. Processing for the failing metric in the current cycle stops, and therefore there are no aggregated health scores for the sub-metric or for the parent metric.

To remediate, you need to resolve the underlying cause of CIs failing the sub-metric tests. See [CMDB Health process status: failure threshold reached](#) for more information about resolving the failures of the sub-metric.

- An error is encountered while processing the sub-metric.

To remediate, examine the system logs to determine the cause of the error. After fixing the cause of the problem, restart processing by manually executing the respective parent metric dashboard job.

CMDB Health reference

Reference topics provide additional information about the KPIs and metrics that CMDB Health uses and about tables, scheduled jobs, and properties that are used by CMDB Health.

- [Domain separation in CMDB Health](#)

This is an overview of domain separation as it pertains to CMDB Health. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains. You can control several aspects of this separation, including which users can see and access data.

Related tasks

- [View relationships health](#)
- [View CI health](#)
- [Create a CMDB remediation rule](#)

Related concepts

- [Exploring CMDB Health](#)
- [CMDB Health experience in CMDB Workspace](#)
- [View CMDB Health Dashboard](#)

Related reference

- [CMDB Health process tracking and troubleshooting](#)

Domain separation in CMDB Health

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Overview

CMDB dashboards should be set up with their own set of rules to best accommodate how the user needs them. CMDB dashboard jobs adhere to those rules to produce reports. These are covered in separate sections below.

How domain separation works in CMDB Health

For dashboards to be the most effective, users should configure the dashboard accordingly. This is done by setting up the orphan, staleness, and inclusion rules to meet their needs, which then affect the reports displayed on the dashboard.

The settings and metrics define different aspects of each application because each domain can be configured differently. These rules are set up in addition to those that are included in the base system. There are different types of owners for different CIs; each domain has its own set of rules.

Metric tests from the global domain propagate to subdomains. However, subdomains can have their own local metric tests which override the global domain tests. Up until the San Diego release, subdomain local

metric tests were applied to the subdomain CIs and also to the global domain CIs (which are visible on subdomains). Global domain CIs that failed metric tests of local subdomains, could have generated large amounts of data due to duplicated data.

Starting with the Tokyo release, CIs in the global domain are evaluated only against metric tests specified in the global domain. In subdomains, local metric tests are applied only to the CIs in that subdomain and are not applied to the global domain CIs (even though the global domain CIs are visible in the subdomain). Health results for CIs in the global domain appear on subdomains and health results on subdomains reflect this new behavior.

Note: Domain separation is on by default, but each domain can be configured as needed.

Health Preferences

Configure these preferences during setup:

1. Global system properties that control CMDB Health – System properties are not domain separated. To learn more see [CMDB Health system properties](#).
2. [CMDB Health Dashboard Jobs](#) – There is a dashboard job for each major KPI, such as Completeness. That job finds the health of the CIs across all the enabled domains. There is only one job run for all domains and jobs themselves are not domain separated.

Users can define the frequency with which they want to run jobs; the report runs for all the domains. The more domains included in the job, the longer the job runs.

3. Health Metrics – These selections are domain-separated and adhere to the established “system overrides” logic of domain separation. Changes are made according to the domain for which the user is logged in. Base system values are defined at the global domain. The overriding domain logic means these values apply for all domains. If users want different values for a domain, they must be logged in to a specific domain and change the property from there. The new property setting applies only to that domain and any domain that inherits this domain. To learn more, see [Health Metrics](#).

Note: Regarding the Completeness, Compliance, and Correctness KPIs: Users can disable this KPI if they don't want to see that as part of the dashboard score. All these settings are domain-separated and the user can define specific properties for the domain.

- a. Active – This setting is the most important because it affects how long the jobs run. The more domains with flags set to Active, the longer the jobs take. It's best to select only those domains you wish to be Active and render the rest as Active = **false**. You can set this in Health Preferences. The default settings for global domain are Active = **true**, but you can modify or disable specific domains the user wants to see in the dashboard. Users should consider the domain hierarchy when changing these values. If there is a large number of domains (>100) the job can take a very long time. To mitigate this, set Active to **false** for all the root domains, thereby disabling all the other domains in the hierarchy. If there is a rule at the top, all child domains inherit that rule.
- b. Failure Threshold, Create Task, Task Assignee Group – All these settings can be set differently for different domains depending on what is needed in each domain.
- c. Exceptions – For Relationship metrics (relationship, duplicate relations, orphan relations, stale relations) the failure threshold setting is not domain separated. The Failure Threshold for the global domain is applied to all domains. For example, even if users were to override the Failure Threshold for a domain, the global domain setting for Threshold is still applied.
- d. Troubleshooting / Implementation detail – These settings are stored in the cmdb_health_metric_pref table, which is domain separated.

CMDB Health-related rules

See CMDB Health-related rules settings at:

- [Required](#)
- [Recommended](#)
- [Orphan](#)

- Staleness

Most of the CMDB Health-related rules are domain separated and provided by the users. Users can define different rules for different domains by logging in to each domain and adding/overriding rules in the CI Class Manager.

1. Completeness

- a. Required fields – These are based on the class schema defined in the platform's [System dictionary](#) and is fixed for all domains. These cannot be changed.
- b. Recommended fields – These are domain separated. The table used is CMDB Recommended Fields [cmdb_recommended_fields], which is domain separated. The user can set these up for different domains.

2. Correctness

- a. Duplicates – Duplicates are based on Identification rules, which are not domain separated, so the same rules apply to all domains.
- b. Orphan – Orphan rules are domain separated; there are different orphan rules for different domains. The table used in the CMDB Health Orphan Rule [cmdb_health_orphan_rule] table and is domain separated.
- c. Staleness – Staleness rules are domain separated. The table used is cmdb_health_staleness_rule. The base system rule (60 days) is set for global domain so is inherited by all domains as the default rule.

3. Compliance

Audit – Audit scores are based on the desired state or scripted audits defined in the compliance module by the user. Audits themselves are domain separated. When audit score evaluation is enabled for a domain, scores become based only on the audits visible in that domain.

Health inclusion rules:

- Health inclusion rules are domain separated. The rules are stored in the `cmdb_health_config` table which is domain separated.
- Each domain can have its specific health inclusion rules and domain-specific rules for each sub-metric.
- When a health inclusion rule is defined globally, all sub-domains inherit the rule according to the domain structure and the rule can be overridden at any domain.
- When a health inclusion rule is defined at the Configuration Item [`cmdb_ci`] class level, all descending classes inherit the rule and the rule can be overridden at any class level.

CMDB Health Dashboard

If the Domain Support — Domain Extensions Installer plugin is activated, then the CMDB Health Dashboard is domain aware:

- The CMDB Health Dashboard aggregates and reports health failures and scores based on user's domain visibility of CIs. If domain visibility lets a user see a CI, then the audit rule in that user's domain applies to that CI, whether the CI is in the user's domain or in a contained domain. If a CI fails health tests from different user domains, then separate failure records are created.
- Users can configure KPI and metric settings specific to the needs in their domain. So different domains can have different settings such as active/inactive, and thresholds.
- A child domain derives its immediate parent's domain health configurations if the child domain does not configure its own. A child domain can override parent's configurations by modifying them.

Health Dashboards (Class View/Service View/Health Group View)

In general, CMDB Health dashboards are domain aware and show data according to the logged-on domain user. If a user is logged into a domain and views a health dashboard:

1. Only scores for enabled metrics in that domain display (based on the Health Preferences Active flag as discussed above).
2. All scores are based on CIs that are visible from the specific domain. (These are regular domain visibility rules: From that domain you can

see CIs in global domain, the specific domain, any child domain of that domain or any domain that gets directly or indirectly contained by that domain.)

3. The dashboard view is based on domain rules defined in domain mapping, as opposed to those provided by the logged-in user. This view overrides any additional domain visibility rules that a logged-in user might have. The admin sets the basic rules, but does not set each individual domain. The admin can give specific users or user groups additional visibility to other domains and the dashboard still does not change. The dashboard strictly follows the domain rules mentioned above, based on the domain hierarchy for the domain in which the user is logged in.
4. As explained in the Health Preferences section, users can define different preference values for any domain which impact the scores reported in the dashboard. Preferences that can impact scores include **Failure Threshold** and **Active**.
5. As explained in the CMDB Health Rules section, the scores reported for the metrics are based on the health rules defined for them (staleness, orphan, recommended, audit, and inclusion rules) which can be defined differently for a specific domain (in the CI Class Manager). Only the required metric and duplicate metric are based on rules that apply in all domains.
6. Service View/ Group View – These reports also largely follow the above points. Typically, these views differ from various views/filters for the Health Report. One is based off business rules, the other is based off CMDB Health groups.

Related concepts

- [Domain separation and Configuration Management Database \(CMDB\)](#)

Components installed with CMDB Health

Several types of components are installed with CMDB Health (included in the com.snc.cmdb plugin), such as tables, properties, and scheduled jobs.

Note: The Application Files table lists the components that are installed with this application. For instructions on how to access this table, see [Find components installed with an application](#).

Properties installed

Property	Description
glide.cmdb.services_hierarchy_limit	Maximum number of service CIs that can appear in the CMDB service dashboard. This limit applies to any child class of the Service Instance [cmdb_ci_service_auto] class. • Type: integer • Default: 10,000 • Range: 0-100,000 • Location: Add to System Properties [sys_properties] table. • Learn more: View CMDB Health Dashboard

Scheduled jobs installed

Scheduled job	Description
CMDB Health Dashboard - Completeness Score Calculation	Script for calculating the completeness KPI of CMDB health.
CMDB Health Dashboard - Compliance Score Calculation	Script for calculating the compliance KPI of CMDB health.
CMDB Health Dashboard - Correctness Score Calculation	Script for calculating the correctness KPI of CMDB health.

Scheduled job	Description
CMDB Health Dashboard - Relationship Score Calculation	Script for calculating the CI relationships KPI of CMDB health.
CMDB Health Dashboard - Relationship Compliance Processor	Script for calculating compliance of relationships with suggested relationships, and with hosting and containment rules.
Add Identifier Fields In Recommended Rules	Sets all criterion attributes from all active identifier entries from all active identification rules, as recommended fields. These added recommended fields are then checked by the CMDB Health Dashboard - Completeness Score Calculation scheduled job when evaluating the recommended health metric.
Remove Identifier Fields In Recommended Rules	Identifies any recommended field that is a criterion attribute in any active identifier entry in any active identification rule. Then removes the recommended setting for that field.

Tables installed

Table	Description
CMDB Health Metric [cmdb_health_metric]	Details such as if a KPI or metric is enabled, maximum failure threshold, and other settings for all CMDB Health KPIs and metrics.
CMDB Health Result [cmdb_health_result]	Results from the most recent CMDB Health processing cycle.

Table	Description
CMDB Health Scorecard [cmdb_health_scorecard]	Current and historic health scores. Status of historic score records is 'Historic', and of latest score records is 'Complete'.
CMDB Health Orphan Rule [cmdb_health_orphan_rule]	Rules for calculating orphan records per class.
CMDB Recommended Fields [cmdb_recommended_fields]	Recommended fields per class.
CMDB Health Metric Status [cmdb_health_metric_status]	Internal table that tracks the status of each KPI and metric that is being processed. Includes status, processing time, and processing start date. State for a KPI or metric changes from 'In Progress' to either of: • Complete • MaxFailures • Daily Processing Time Out Processing of a timed out KPI or metric continues on the following day.
CMDB Health Processor Status [cmdb_health_processor_status]	Internal table that tracks the processing progress of each KPI and metric. Contains a list of tables that are processed for each KPI and metric, and processing status. Classes are processed sequentially, changing status from Draft -> In Progress -> Complete.

Table	Description
CMDB Relationship All Rules Health Results [cmdb_health_result_rel_all]	Stores results about relationship health, to be used by the All Relationships report.